

INDEX METHODOLOGY

Black Scarab Physical AI 15

Rules for composition, calculation, and maintenance

BSPI15

Version 1.0 | Effective 1 October 2026
Black Scarab Intelligence



Methodology status

This document records the operating rules used for the composition, calculation, and maintenance of the Black Scarab Physical AI 15.

Field	Value
Index name	Black Scarab Physical AI 15
Ticker	BSPI15
Version	1.0
Base date	1 October 2026
Base level	1,000.00
Return type	Currency neutral price return
Review frequency	Quarterly
Publisher	Black Scarab Intelligence

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An adoption focused research benchmark

BSPI15 measures the local market performance of fifteen listed companies supplying systems and enabling technologies used in physical automation. The index spans robot platforms, motion and actuation, perception and navigation, control and edge compute, and deployment and autonomy.

The index is designed as a transparent research benchmark. It is not an investment fund, security, financial product, investment recommendation, or offer to buy or sell securities.

Design principles

Physical AI relevance A constituent must have material operating exposure to embodied intelligence, robotics, autonomous physical systems, or directly specialized enabling technology.	Global opportunity set Eligible issuers may be domiciled or listed in any country when the listing and data requirements are satisfied.
Commercial breadth Selection favors established products, diversified customers, and repeat deployments over isolated pilots or a single program.	Concentration control Equal weighting limits the influence of the largest issuers and keeps component suppliers visible in the index result.

Interpretation

The methodology should be read as a complete document. Definitions apply throughout. When an event is not expressly covered, the publisher may determine a treatment that preserves continuity and the economic intent of the index. Any such decision must be recorded in the index maintenance log.

Core characteristics

Characteristic	Rule
Asset class	Listed equity securities
Regional allocation	Global
Number of constituents	15
Weighting	Equal weight at each quarterly review
Return treatment	Price return. Cash dividends are excluded.
Currency treatment	Local listing returns are combined without foreign exchange conversion
Calculation frequency	One official closing level for each index calculation day
Precision	Published index level rounded to two decimal places

Publication

The official closing series, current constituents, effective weights, and methodology are published through blackscarab.ai. Historical files and provider receipts are retained in the index data archive.

Benchmark comparisons

The website may compare BSP15 with the S&P 500 Price Return Index and Nasdaq Composite Price Return Index, rebased to the BSP15 base level on the base date. Benchmark comparisons do not alter the index calculation.

Eligibility requirements

- The security represents an operating company listed on a recognized public exchange.
- Daily regular market closing prices are observable for the selected primary listing.
- The issuer provides public financial and operating information sufficient to assess its business exposure.
- The company has material exposure to at least one BSPI15 operating sleeve.
- The relevant exposure is supported by operating products, deployed systems, recurring commercial activity, or a clearly identifiable specialized business unit.

Exclusions

- Private companies, funds, special purpose vehicles, and securities without a usable closing price history.
- Companies whose connection to Physical AI is incidental, customer based, partner based, or limited to a single exploratory project.
- Duplicate share classes when one primary listing sufficiently represents the issuer.

Selection process

At each quarterly review, the publisher evaluates the eligible universe using public filings, company materials, exchange information, and credible independent reporting. Candidates are assessed for relevance, commercial breadth, product maturity, customer diversification, and the durability of their role in the Physical AI supply chain.

The fifteen selected securities should collectively cover the index operating sleeves. The sleeve labels describe economic exposure. They are not separate subindices and do not impose fixed sleeve weights.

Subsidiaries and parent companies

A separately listed subsidiary may be selected when its own operating purpose satisfies the methodology. A parent company is not included solely because it owns or invests in a relevant subsidiary. Only one security is selected for the same economic exposure unless the businesses are separately listed and independently material.

Five operating sleeves

Robot platforms Industrial robots, collaborative systems, and autonomous mobile platforms.	Motion and actuation Servo systems, drives, controls, gearing, and components that create physical movement.
Perception and navigation Machine vision, lidar, sensing, identification, mapping, and robotic guidance.	Control and edge compute Real time control, industrial networking, edge processing, and onboard intelligence.
Deployment and autonomy Material handling, positioning, integration, and operating systems used in scaled deployments.	

Primary classification

Each constituent receives one primary sleeve based on the activity that most directly explains its role in Physical AI adoption. A company may have additional relevant activities, but it is counted once.

Target weights

Each constituent receives a target weight of 1/15, or approximately 6.6667 percent, at each quarterly rebalance. No country, currency, sleeve, or market capitalization overlay is applied.

Weight drift

Between rebalances, effective weights change with constituent price performance. Published current weights therefore differ from target weights. The target allocation view remains fixed until the next rebalance.

Local currency price return

For constituent i on calculation day t , the local price return is:

$$r_{i,t} = P_{i,t} / P_{i,t-1} - 1$$

where P is the unadjusted regular market closing price in the currency of the selected listing. Foreign exchange conversion is not applied.

The index closing level is:

$$I_t = I_{t-1} \times (1 + \sum w_{i,t-1} r_{i,t})$$

where w is the constituent effective weight immediately before the current return is applied.

Effective weight

Between rebalances, each weight is proportional to the constituent price relative since the last rebalance. The effective weights sum to 100 percent.

Dividends

Ordinary cash dividends are excluded. The index is therefore a price return series, not a net total return or gross total return series.

Non trading days

If a constituent exchange is closed on an index calculation day, the most recent valid closing price is carried forward and the constituent return for that index day is zero.

Quarterly review

The index is reviewed after the final trading day of March, June, September, and December. Approved constituent changes and target weights take effect on the following common trading day for which the required implementation prices are available.

Review record

- Eligible universe and relevance assessment.
- Selected constituents, primary listings, tickers, currencies, and sleeve classifications.
- Additions, deletions, and reasons for each change.
- Target weights and the prices used to determine effective index shares.

Extraordinary events

A constituent may require action before a scheduled review following a delisting, insolvency, acquisition, trading suspension, or another event that prevents a representative closing price. The publisher determines whether to remove, replace, or carry the constituent based on continuity, available evidence, and the next practical implementation date. The decision and effective date are recorded before the canonical index file is changed.

Corporate actions

Share splits, reverse splits, stock distributions, mergers, spin offs, rights issues, and similar events are handled through price or share adjustments intended to prevent the mechanical event itself from changing the index level. Cash dividends remain excluded.

Daily control framework

- Map every constituent to an approved provider symbol and expected exchange MIC.
- Retrieve unadjusted closing price, adjusted close, dividend, and split fields while using only the unadjusted close for the index return.
- Compare every available source observation with the canonical close.
- Retain immutable provider responses and a dated observation ledger.
- Stop the production update when a required observation is delayed, missing, or inconsistent unless an exchange holiday is independently confirmed.

Source hierarchy

Marketstack is the approved primary daily market data source. Exchange notices, issuer materials, and other authoritative sources are used to verify market holidays, ticker changes, and corporate actions.

Corrections

A confirmed data or calculation error is corrected in the canonical history and documented with the affected date, original value, corrected value, evidence, and approval date. A correction is not used to rewrite a valid historical methodology decision.

Delayed observations

A preliminary review may identify provider observations that have not finalized. The public closing series is updated only after the required prices pass the source and identity checks.

Publisher responsibilities

Black Scarab Intelligence maintains the methodology, constituent record, calculation files, daily source receipts, and published index pages. Material decisions must be evidence based and recorded.

Methodology review

The methodology is reviewed at least annually and may be reviewed earlier when market structure, data availability, or the Physical AI industry changes materially.

Methodology changes

A material change is documented in the version history before it becomes effective. The notice describes the rule being changed, the reason, the effective date, and any effect on historical comparability. Historical values are not restated solely because a future rule changes.

Discretion

When expert judgment is required, the publisher applies it consistently with the index objective, available evidence, continuity, and equal treatment of comparable cases. The decision is retained in the maintenance record.

Termination

The index may be discontinued if reliable calculation becomes impractical, the eligible universe no longer supports the stated objective, or continued publication is no longer sustainable. A termination notice should identify the final calculation date and the reason.

Defined terms

Calculation day A weekday on which the index can be calculated using required closing prices or valid carried forward prices.	Closing price The final regular market price published for the selected listing, or the most recent valid price when the exchange is closed.
Constituent A listed security included in BSP115.	Effective weight A constituent's share of the index immediately before the current day's return is applied.
Primary listing The exchange listing selected as the canonical security and price source for the issuer.	Price relative Current closing price divided by the rebalance reference price, adjusted when required for capital events.
Rebalance The scheduled reset of constituents and target weights.	Review date The date on which eligibility, selection, classification, and target weights are assessed.

Interpretation of time and dates

Dates refer to the local calendar of the relevant exchange unless the context states otherwise. A common trading day is a day on which the implementation prices required for all affected constituents are available.

Initial constituents

Company	Ticker	Exchange	Country	Primary sleeve	Target
FANUC	6954	Tokyo	Japan	Robot platforms	6.67%
Yaskawa Electric	6506	Tokyo	Japan	Robot platforms	6.67%
Estun Automation	002747	Shenzhen	China	Robot platforms	6.67%
Teradyne	TER	Nasdaq	United States	Robot platforms	6.67%
Nabtesco	6268	Tokyo	Japan	Motion and actuation	6.67%
Inovance	300124	Shenzhen	China	Motion and actuation	6.67%
Keyence	6861	Tokyo	Japan	Perception and navigation	6.67%
Cognex	CGNX	Nasdaq	United States	Perception and navigation	6.67%
Ouster	OUST	NYSE	United States	Perception and navigation	6.67%
Hesai	HSAL	Nasdaq	China	Perception and navigation	6.67%
Rockwell Automation	ROK	NYSE	United States	Control and edge compute	6.67%
Siemens	SIE	Xetra	Germany	Control and edge compute	6.67%
NXP Semiconductors	NXPI	Nasdaq	Netherlands	Control and edge compute	6.67%
Daifuku	6383	Tokyo	Japan	Deployment and autonomy	6.67%
Hexagon	HEXA B	Stockholm	Sweden	Deployment and autonomy	6.67%

The initial constituent set became effective on 1 October 2026. Current effective weights drift with price performance and are published separately.

Version history

Version	Date	Status	Summary
1.0	1 October 2026	Effective	Initial documented methodology for BSP15.

Important information

BSP15 is an editorial research index published for informational purposes. It is not administered as a regulated benchmark and is not currently licensed as the basis of an investment product. Nothing in this document is investment, legal, tax, or accounting advice.